GRADE 7 ACCELERATED · UNIT 4 · ASSESSMENT

# Solving Problems with Rational Numbers

*Form A · 29 points · one class period* **ANSWER KEY**

Name
Date
Period

*Show your work. One point for each lettered part.*

## Part 1. Operations in Scientific Notation

*12 points MA.8.NSO.1.5*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**1.** Add or subtract. Write each answer in scientific notation.

**a.** $(3.6 \times 10^5) + (2.1 \times 10^5)$ *1 pt*

$5.7 \times 10^5$

*Same power: add the coefficients.*

**b.** $(8.4 \times 10^{-6}) - (5.9 \times 10^{-6})$ *1 pt*

$2.5 \times 10^{-6}$

**c.** $(7.5 \times 10^8) + (6.8 \times 10^8)$ *1 pt*

$1.43 \times 10^9$

$14.3 \times 10^8$ *is the value; 14.3 is not less than 10, so it is not the answer.*

**d.** $(4.2 \times 10^4) + (3.7 \times 10^3)$ *1 pt*

$4.57 \times 10^4$

$3.7 \times 10^3 = 0.37 \times 10^4$ .

*Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.*

**2.** Multiply or divide. Write each answer in scientific notation.

**a.** $(3 \times 10^4)(2 \times 10^6)$ *1 pt*

$6 \times 10^{10}$

**b.** $\frac{9.6 \times 10^9}{1.2 \times 10^4}$ *1 pt*

$8 \times 10^5$

**c.** $(5 \times 10^{-3})(4 \times 10^7)$ *1 pt*

$2 \times 10^5$

$20 \times 10^4$ *is not finished.*

**d.** $\frac{1.8 \times 10^2}{6 \times 10^5}$ *1 pt*

$3 \times 10^{-4}$

$0.3 \times 10^{-3}$ *is not finished either.*

*Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.*

**3.** Which is the value of $(1.4 \times 10^3) + (2.9 \times 10^5)$ ? *1 pt*

- ○ A. $3.04 \times 10^5$
- ○ B. $4.3 \times 10^5$
- ● **C.** $2.914 \times 10^5$
- ○ D. $4.3 \times 10^8$

***C***

*Scoring: 1 point.*

**4.** Select ALL of the expressions whose value is $3 \times 10^6$ . *1 pt*

- ☒ **A.** $(3.2 \times 10^6) - (2 \times 10^5)$
- ☐ B. $\frac{9 \times 10^9}{3 \times 10^4}$
- ☒ **C.** $(6 \times 10^3)(5 \times 10^2)$
- ☒ **D.** $(1.8 \times 10^6) + (1.2 \times 10^6)$
- ☐ E. $(1.8 \times 10^6) + (1.2 \times 10^5)$
- ☒ **F.** $\frac{9 \times 10^9}{3 \times 10^3}$

***A, C, D and F***

*C gives* $30 \times 10^5$ *, which is* $3 \times 10^6$ *once it is finished. All four correct and nothing else for the point.*

*Scoring: 1 point.*

**5.** Kendra writes $(2.3 \times 10^4) + (6.1 \times 10^6) = 8.4 \times 10^{10}$. Name the two mistakes she made, and give the correct sum. *1 pt*

***She added the coefficients without first writing both numbers over the same power of ten, and she added the exponents. The correct sum is*** $6.123 \times 10^6$ ***.***

*Both mistakes must be named for the point; one of the two is half the item.*

$2.3 \times 10^4 = 0.023 \times 10^6$ .

*Scoring: 1 point.*

**6.** A number is multiplied by $4 \times 10^3$ and the product is $2 \times 10^{-1}$. What was the number? Write your answer in scientific notation. *1 pt*

$5 \times 10^{-5}$

*TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed.*

$\frac{2 \times 10^{-1}}{4 \times 10^3} = 0.5 \times 10^{-4} = 5 \times 10^{-5}$ *. Working a product backwards across a sign change appears on no review and in no question bank; the point is for the finished form, so* $0.5 \times 10^{-4}$ *alone does not earn it.*

*Scoring: 1 point.*

## Part 2. Real-World Problems in Scientific Notation

*4 points MA.8.NSO.1.6*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**7.** Solve. Write each answer in scientific notation. Do not round.

**a.** A rectangular plate measures $2.4 \times 10^2$ mm by $1.35 \times 10^2$ mm. Find its area. *1 pt*

$3.24 \times 10^4$ ***square millimeters***

*Multiply the coefficients and add the exponents: 2.4 · 1.35 = 3.24, and* $10^2 \cdot 10^2 = 10^4$ .

**b.** A vehicle travels $5.6 \times 10^2$ km in 8 hours. Find its average speed. *1 pt*

$7 \times 10^1$ ***km/h***

$\frac{5.6 \times 10^2}{8} = 0.7 \times 10^2$ *, which is not finished: 0.7 is less than 1.*

*Scoring: a 1 · b 1 = 2 points.*

**8.** Solve. Write each answer in scientific notation. Do not round.

**a.** Two lengths are measured: $6.25 \times 10^3$ mm and $4.1 \times 10^2$ mm. Find their sum. *1 pt*

$6.66 \times 10^3$ ***mm***

*Match the powers first:* $4.1 \times 10^2 = 0.41 \times 10^3$ *, and 6.25 + 0.41 = 6.66.*

**b.** Two masses are measured: $9.3 \times 10^{-2}$ g and $4.72 \times 10^{-3}$ g. Find their difference. *1 pt*

$8.828 \times 10^{-2}$ ***g***

*Match the powers first:* $4.72 \times 10^{-3} = 0.472 \times 10^{-2}$ *, and 9.3 − 0.472 = 8.828.*

*Scoring: a 1 · b 1 = 2 points.*

## Part 3. The Order of Operations with Exponents and Radicals *13 points*

*MA.8.NSO.1.7*

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.*

**9.** Evaluate each expression.

**a.** $15 - 2 \cdot 3^2$ *1 pt*

***−3***

*The power first, then the product, then the subtraction.*

**b.** $\dfrac{5^2 + 7}{2^3}$ *1 pt*

***4***

*32 over 8; the bar groups the whole numerator.*

**c.** $36 \div 4 \cdot 3$ *1 pt*

***27***

*One level, left to right: 9 · 3. Not 3.*

**d.** $-4^2 + (-4)^2$ *1 pt*

***0***

*−16 + 16. The parentheses decide what the exponent is attached to.*

*Scoring: a 1 · b 1 · c 1 · d 1 = 4 points.*

**10.** Evaluate each expression.

**a.** $\sqrt{6^2 + 13}$ *1 pt*

***7***

*Finish under the bar: 36 + 13 = 49.*

**b.** $\sqrt[3]{-125} + 2^{-2}$ *1 pt*

$-\frac{19}{4}$

$-5 + \frac{1}{4}$ .

**c.** $\left(-\frac{1}{3}\right)^2 + \sqrt{\frac{4}{9}}$ *1 pt*

$\frac{7}{9}$

$\frac{1}{9} + \frac{2}{3}$ .

*Scoring: a 1 · b 1 · c 1 = 3 points.*

**11.** A square patio of side 14 feet is sealed at 2.40 dollars a square foot, with a flat 85 dollars for delivery.

**a.** Write an expression for the total cost. *1 pt*

$14^2(2.40) + 85$

*The rate multiplies the area; the flat fee does not. An expression that charges 85 on every square foot does not earn the point.*

**b.** Find the total cost. *1 pt*

***555.40 dollars***

*196(2.40) = 470.40, then + 85.*

*Scoring: a 1 · b 1 = 2 points.*

**12.** Which is the value of $2 + 5 \cdot 3^2$ ? *1 pt*

- ○ A. 32
- ● **B. 47**
- ○ C. 63
- ○ D. 289

***B***

*Scoring: 1 point.*

**13.** Select ALL of the expressions whose value is 6. *1 pt*

- ☐ A. $-\sqrt{36}$
- ☒ **B.** $\sqrt{36}$
- ☐ C. $36 \div 3 \cdot 2$
- ☒ **D.** $\left(\frac{1}{6}\right)^{-1}$
- ☒ **E.** $\sqrt{5^2 + 11}$
- ☒ **F.** $\frac{4^2 + 8}{4}$

***B, D, E and F***

*All four correct and nothing else for the point.*

*Scoring: 1 point.*

**14.** Nadia writes $\sqrt{25 + 144} = \sqrt{25} + \sqrt{144} = 17$ . Explain why she is wrong, and give the correct value. *1 pt*

***The radical bar is a grouping symbol, so 25 + 144 is finished before the root is taken:*** $\sqrt{169} = 13$ ***. The root of a sum is not the sum of the roots.***

*The point needs the reason AND the value. '13' alone does not earn it, and neither does 'she added wrong'.*

*Scoring: 1 point.*

**15.** A square tile has an area of 169 square inches. A second square tile has a side twice as long as the first. What is the area of the second tile? *1 pt*

***676 square inches***

*TRANSFER ITEM — not on the practice test. Same benchmark, a surface nobody rehearsed. The side of the first is* $\sqrt{169} = 13$ *, so the second has side 26 and area* $26^2 = 676$ *. Doubling a side and asking for the new area appears on no review and in no question bank; a student who doubles the area and writes 338 has not gone through the side.*

*Scoring: 1 point.*

## Score Tracker

| Benchmark | Questions | Possible | Earned |
|---|---|---|---|
| MA.8.NSO.1.5 | 1, 2, 3, 4, 5, 6 | 12 | |
| MA.8.NSO.1.6 | 7, 8 | 4 | |
| MA.8.NSO.1.7 | 9, 10, 11, 12, 13, 14, 15 | 13 | |
| **Total** | | **29** | |

*If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is*

*arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.*